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JEE MAIN · QUESTION PAPER

JEE Main 2022 (Online) 24th June Morning Shift

JEE Main Question Paper with Answer Key

24 Jun 2022	90	21	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	21 questions
Mathematics	22 questions
Physics	26 questions
Out of syllabus	21 questions
Answer key	at the end

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Chemistry 21 questions

Q1. MCQ (Single Correct Answer)

If a rocket runs on a fuel ($C_{15}H_{30}$) and liquid oxygen, the weight of oxygen required and CO_2 released for every litre of fuel respectively are :

(Given : density of the fuel is 0.756 g/mL)

Open solution

(A) 1188 g and 1296 g

(B) 2376 g and 2592 g

(C) 2592 g and 2376 g

(D) 3429 g and 3142 g

Q2. MCQ (Single Correct Answer)

Consider the following pairs of electrons

(A) (a) n = 3, l = 1, m_l = 1, m_s = + $\frac{1}{2}$

(b) n = 3, l = 2, m_l = 1, m_s = + $\frac{1}{2}$

(B) (a) n = 3, l = 2, m_l = -2, m_s = $-\frac{1}{2}$

(b) n = 3, l = 2, m_l = -1, m_s = $-\frac{1}{2}$

(C) (a) n = 4, l = 2, m_l = 2, m_s = + $\frac{1}{2}$

(b) n = 3, l = 2, m_l = 2, m_s = + $\frac{1}{2}$

The pairs of electrons present in degenerate orbitals is/are :

Open solution

(A) Only (A) **(B)** Only (B) **(C)** Only (C) **(D)** (B) and (C)

Q3. MCQ (Single Correct Answer)

Match List - I with List - II :

	List - I		List -II
(A)	$[PtCl_4]^{2-}$	(I)	sp^3d
(B)	BrF_5	(II)	d^2sp^3
(C)	PCl_5	(III)	dsp^2
(D)	$[Co(NH_3)_6]^{3+}$	(IV)	sp^3d^2

Open solution

Choose the most appropriate answer from the options given below :

(A) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

(B) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

(C) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(D) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

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Q4. MCQ (Single Correct Answer)

Open solution

For a reaction at equilibrium

$A(g) \rightleftharpoons B(g) + \frac{1}{2} C(g)$

the relation between dissociation constant (K), degree of dissociation (α) and equilibrium pressure (p) is given by :

(A) $K = \frac{\alpha^{\frac{1}{2}} p^{\frac{3}{2}}}{\left(1+\frac{3}{2}\alpha\right)^{\frac{1}{2}}(1-\alpha)}$

(B) $K = \frac{\alpha^{\frac{3}{2}} p^{\frac{1}{2}}}{(2+\alpha)^{\frac{1}{2}}(1-\alpha)}$

(C) $K = \frac{(\alpha p)^{\frac{3}{2}}}{\left(1+\frac{3}{2}\alpha\right)^{\frac{1}{2}}(1-\alpha)}$

(D) $K = \frac{(\alpha p)^{\frac{3}{2}}}{(1+\alpha)(1-\alpha)^{\frac{1}{2}}}$

Q5. MCQ (Single Correct Answer)

Open solution

Given below are the oxides:

Na_2O, As_2O_3, N_2O, NO and Cl_2O_7

Number of amphoteric oxides is :

(A) 0 (B) 1 (C) 2 (D) 3

Q6. MCQ (Single Correct Answer)

Open solution

The most stable trihalide of nitrogen is :

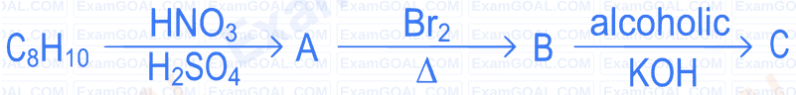
(A) NF_3 (B) NCl_3 (C) NBr_3 (D) NI_3

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Q7. MCQ (Single Correct Answer)

Open solution

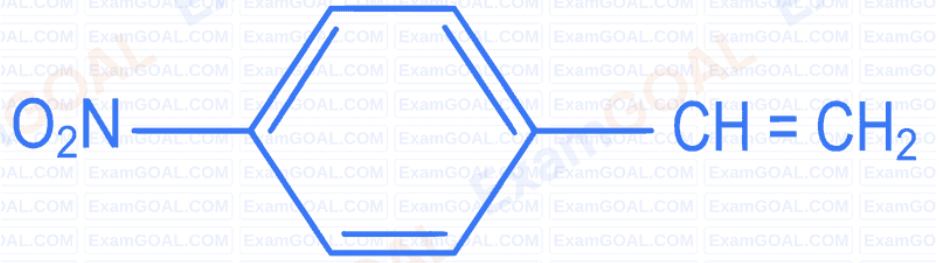
In the given reaction sequence, the major product 'C' is :



(A)



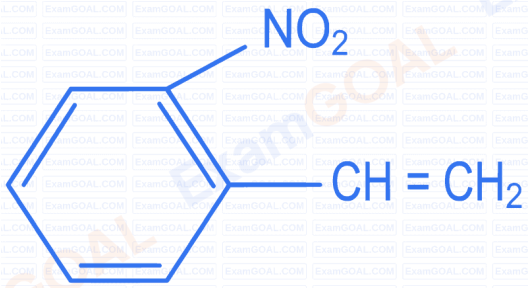
(B)



(C)



(D)



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Q8. MCQ (Single Correct Answer)

Open solution

Two statements are given below :

Statement I : The melting point of monocarboxylic acid with even number of carbon atoms is higher than that of with odd number of carbon atoms acid immediately below and above it in the series.

Statement II : The solubility of monocarboxylic acids in water decreases with increase in molar mass.

Choose the most appropriate option :

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

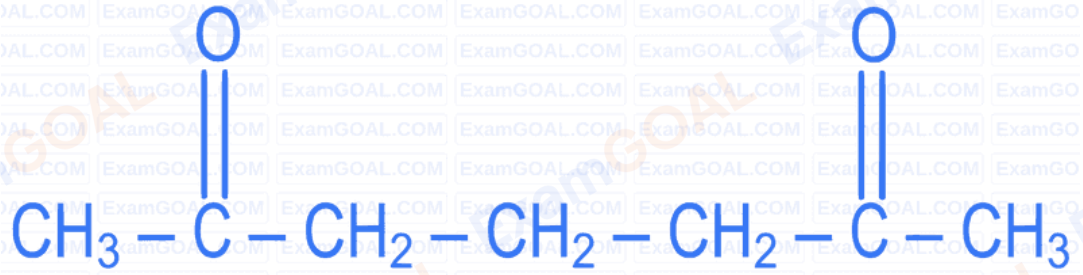
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Q9. MCQ (Single Correct Answer)

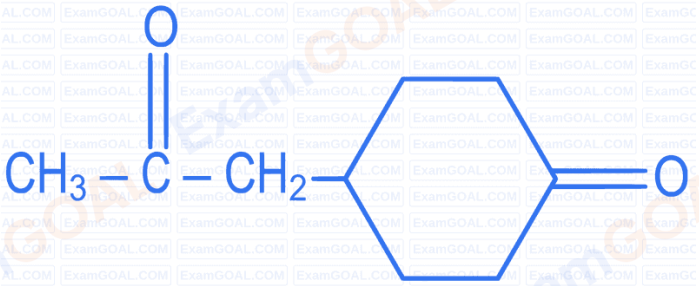
Open solution

Which of the following is an example of conjugated diketone?

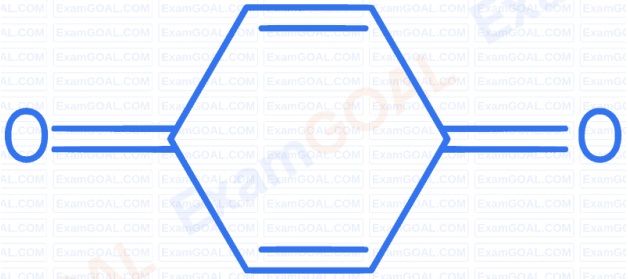
(A)



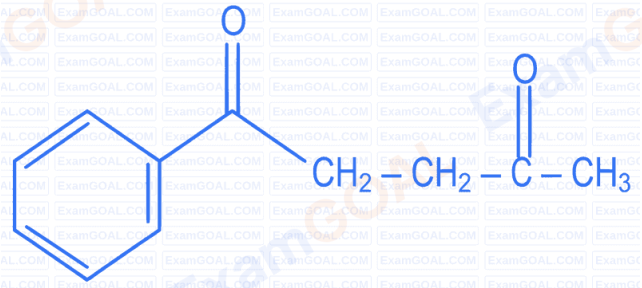
(B)



(C)



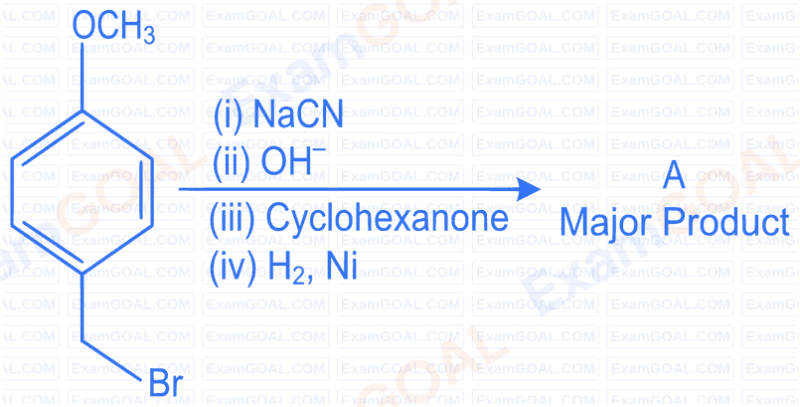
(D)



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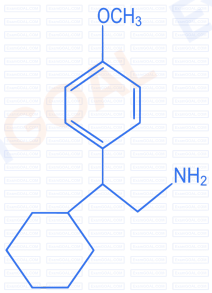
Q10. MCQ (Single Correct Answer)

Open solution

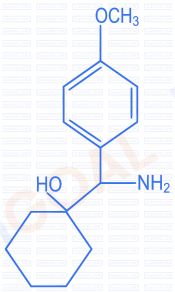


The major product of the above reactions is :

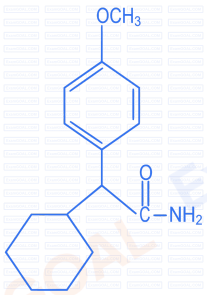
(A)



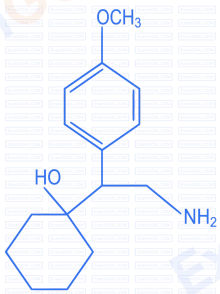
(B)



(C)



(D)



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Q11. MCQ (Single Correct Answer)

A polysaccharide 'X' on boiling with dil H_2SO_4 at 393 K under 2-3 atm pressure yields 'Y'. 'Y' on treatment with bromine water gives gluconic acid. 'X' contains β-glycosidic linkages only. Compound 'X' is :

Open solution

(A) starch (B) cellulose (C) amylose (D) amylopectin

Q12. MCQ (Single Correct Answer)

During the qualitative analysis of salt with cation y^{2+}, addition of a reagent (X) to alkaline solution of the salt gives a bright red precipitate. The reagent (X) and the cation (y^{2+}) present respectively are :

Open solution

(A) Dimethylglyoxime and Ni^{2+}

(B) Dimethylglyoxime and Co^{2+}

(C) Nessler's reagent and Hg^{2+}

(D) Nessler's reagent and Ni^{2+}

Q13. Numerical

$2O_3(g) \rightleftharpoons 3O_2(g)$

At 300 K, ozone is fifty percent dissociated. The standard free energy change at this temperature and 1 atm pressure is $(-)$ _____ J mol^{-1}. (Nearest integer)

[Given : ln 1.35 = 0.3 and R = 8.3 J K^{-1} mol^{-1}]

Open solution

Q14. Numerical

The osmotic pressure of blood is 7.47 bar at 300 K. To inject glucose to a patient intravenously, it has to be isotonic with blood. The concentration of glucose solution in gL^{-1} is _____.

(Molar mass of glucose = 180 g mol^{-1}, R = 0.083 L bar K^{-1} mol^{-1}) (Nearest integer)

Open solution

Q15. Numerical

The cell potential for the following cell

Pt $|H_2(g)|H^+$ (aq)|| Cu^{2+} (0.01 M)|Cu(s)

is 0.576 V at 298 K. The pH of the solution is _____. (Nearest integer)

(Given : $E^o_{Cu^{2+}/Cu} = 0.34$ V and $\frac{2.303\,RT}{F} = 0.06$ V)

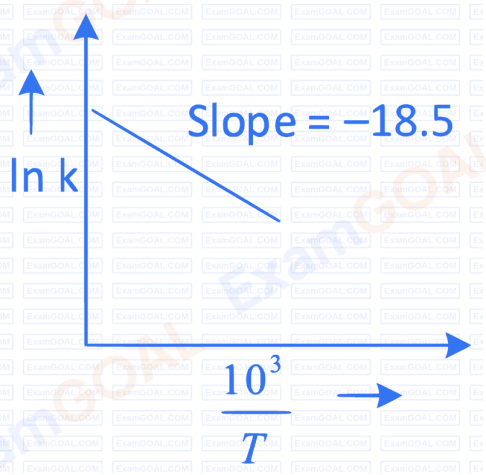
Open solution

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Q16. Numerical

The rate constants for decomposition of acetaldehyde have been measured over the temperature range 700 - 1000 K. The data has been analysed by plotting ln k vs $\frac{10^3}{T}$ graph. The value of activation energy for the reaction is _____ kJ mol^{-1}. (Nearest integer)

(Given : R = 8.31 J K^{-1} mol^{-1})



Open solution

Q17. Numerical

The difference in oxidation state of chromium in chromate and dichromate salts is _____.

Open solution

Q18. Numerical

In the cobalt-carbonyl complex : $[Co_2(CO)_8]$, number of Co-Co bonds is "X" and terminal CO ligands is "Y". X + Y = _____.

Open solution

Q19. Numerical

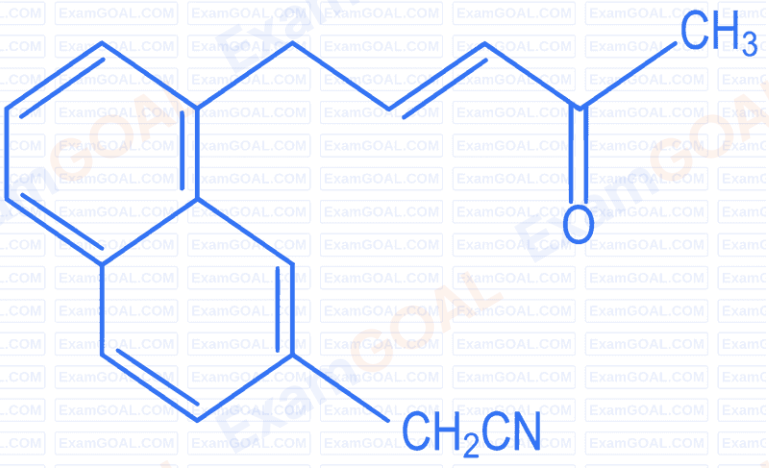
A 0.166 g sample of an organic compound was digested with conc. H_2SO_4 and then distilled with NaOH. The ammonia gas evolved was passed through 50.0 mL of 0.5 N H_2SO_4. The used acid required 30.0 mL of 0.25 N NaOH for complete neutralization. The mass percentage of nitrogen in the organic compound is _____.

Open solution

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Q20. Numerical

Number of electrophilic centres in the given compound is _____.



Open solution

Q21. Numerical

The major product 'A' of the following given reaction has _____ sp^2 hybridized carbon atoms.



Open solution

Mathematics 22 questions

Q22. MCQ (Single Correct Answer)

Let $A = \{z \in C : 1 \leq |z - (1+i)| \leq 2\}$

and $B = \{z \in A : |z - (1-i)| = 1\}$. Then, B :

(A) is an empty set

(B) contains exactly two elements

(C) contains exactly three elements

(D) is an infinite set

Open solution

Q23. MCQ (Single Correct Answer)

The remainder when 3^{2022} is divided by 5 is :

(A) 1 (B) 2 (C) 3 (D) 4

Open solution

Q24. MCQ (Single Correct Answer)

The surface area of a balloon of spherical shape being inflated, increases at a constant rate. If initially, the radius of balloon is 3 units and after 5 seconds, it becomes 7 units, then its radius after 9 seconds is :

(A) 9 (B) 10 (C) 11 (D) 12

Open solution

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Q25. MCQ (Single Correct Answer)

Bag A contains 2 white, 1 black and 3 red balls and bag B contains 3 black, 2 red and n white balls. One bag is chosen at random and 2 balls drawn from it at random, are found to be 1 red and 1 black. If the probability that both balls come from Bag A is $\frac{6}{11}$, then n is equal to _____.

Open solution

(A) 13 (B) 6 (C) 4 (D) 3

Q26. MCQ (Single Correct Answer)

The number of values of α for which the system of equations :

$x + y + z = \alpha$

$\alpha x + 2\alpha y + 3z = -1$

$x + 3\alpha y + 5z = 4$

is inconsistent, is

Open solution

(A) 0 (B) 1 (C) 2 (D) 3

Q27. MCQ (Single Correct Answer)

If the sum of the squares of the reciprocals of the roots α and β of

the equation $3x^2 + \lambda x - 1 = 0$ is 15, then $6(\alpha^3 + \beta^3)^2$ is equal to :

Open solution

(A) 18 (B) 24 (C) 36 (D) 96

Q28. MCQ (Single Correct Answer)

The set of all values of k for which

$(\tan^{-1}x)^3 + (\cot^{-1}x)^3 = k\pi^3,\ x \in R$, is the interval :

Open solution

(A) $\left[\frac{1}{32}, \frac{7}{8}\right)$ (B) $\left(\frac{1}{24}, \frac{13}{16}\right)$

(C) $\left[\frac{1}{48}, \frac{13}{16}\right]$ (D) $\left[\frac{1}{32}, \frac{9}{8}\right)$

Q29. MCQ (Single Correct Answer)

For the function

$f(x) = 4\log_e(x-1) - 2x^2 + 4x + 5,\ x > 1$, which one of the following is NOT correct?

Open solution

(A) f is increasing in (1, 2) and decreasing in $(2, \infty)$

(B) f(x) = -1 has exactly two solutions

(C) $f'(e) - f''(2) < 0$

(D) f(x) = 0 has a root in the interval (e, e + 1)

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Q30. MCQ (Single Correct Answer)

The sum of absolute maximum and absolute minimum values of the function $f(x) = |2x^2 + 3x - 2| + \sin x \cos x$ in the interval [0, 1] is :

(A) $3 + \frac{\sin(1)\cos^2\left(\frac{1}{2}\right)}{2}$

(B) $3 + \frac{1}{2}(1 + 2\cos(1))\sin(1)$

(C) $5 + \frac{1}{2}(\sin(1) + \sin(2))$

(D) $2 + \sin\left(\frac{1}{2}\right)\cos\left(\frac{1}{2}\right)$

Open solution

Q31. MCQ (Single Correct Answer)

If $\{a_i\}_{i=1}^{n}$, where n is an even integer, is an arithmetic progression with common difference 1, and $\sum_{i=1}^{n} a_i = 192$, $\sum_{i=1}^{n/2} a_{2i} = 120$, then n is equal to :

(A) 48 (B) 96 (C) 92 (D) 104

Open solution

Q32. MCQ (Single Correct Answer)

If x = x(y) is the solution of the differential equation

$y\frac{dx}{dy} = 2x + y^3(y+1)e^y$, $x(1) = 0$; then x(e) is equal to :

(A) $e^3(e^e - 1)$ (B) $e^e(e^3 - 1)$ (C) $e^2(e^e + 1)$ (D) $e^e(e^2 - 1)$

Open solution

Q33. MCQ (Single Correct Answer)

Let $\hat{a}$, $\hat{b}$ be unit vectors. If $\vec{c}$ be a vector such that the angle between $\hat{a}$ and $\vec{c}$ is $\frac{\pi}{12}$, and $\hat{b} = \vec{c} + 2\left(\vec{c} \times \hat{a}\right)$, then $\left|6\vec{c}\right|^2$ is equal to :

(A) $6\left(3 - \sqrt{3}\right)$ (B) $3 + \sqrt{3}$ (C) $6\left(3 + \sqrt{3}\right)$ (D) $6\left(\sqrt{3} + 1\right)$

Open solution

Q34. MCQ (Single Correct Answer)

The domain of the function

$f(x) = \frac{\cos^{-1}\left(\frac{x^2-5x+6}{x^2-9}\right)}{\log_e(x^2-3x+2)}$ is :

(A) $(-\infty, 1) \cup (2, \infty)$

(B) $(2, \infty)$

(C) $\left[-\frac{1}{2}, 1\right) \cup (2, \infty)$

(D) $\left[-\frac{1}{2}, 1\right) \cup (2, \infty) - \left\{3, \frac{3+\sqrt{5}}{2}, \frac{3-\sqrt{5}}{2}\right\}$

Open solution

Q35. Numerical

The number of one-one functions f : {a, b, c, d} $\rightarrow$ {0, 1, 2, ..., 10} such

that 2f(a) $-$ f(b) + 3f(c) + f(d) = 0 is _____.

Open solution

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Q36. Numerical

In an examination, there are 5 multiple choice questions with 3 choices, out of which exactly one is correct. There are 3 marks for each correct answer, -2 marks for each wrong answer and 0 mark if the question is not attempted. Then, the number of ways a student appearing in the examination gets 5 marks is _____.

Open solution

Q37. Numerical

Let $A\left(\frac{3}{\sqrt{a}}, \sqrt{a}\right)$, $a > 0$, be a fixed point in the xy-plane. The image of A in y-axis be B and the image of B in x-axis be C. If $D(3\cos\theta, a\sin\theta)$ is a point in the fourth quadrant such that the maximum area of ΔACD is 12 square units, then a is equal to _____.

Open solution

Q38. Numerical

Let a line having direction ratios, 1, -4, 2 intersect the lines $\frac{x-7}{3} = \frac{y-1}{-1} = \frac{z+2}{1}$ and $\frac{x}{2} = \frac{y-7}{3} = \frac{z}{1}$ at the points A and B. Then $(AB)^2$ is equal to _____.

Open solution

Q39. Numerical

The number of points where the function

$$f(x) = \begin{cases} |2x^2 - 3x - 7| & if \quad x \le -1 \\ [4x^2 - 1] & if \quad -1 < x < 1 \\ |x+1| + |x-2| & if \quad x \ge 1 \end{cases}$$

[t] denotes the greatest integer $\le$ t, is discontinuous is _____.

Open solution

Q40. Numerical

Let $f(\theta) = \sin\theta + \int\limits_{-\pi/2}^{\pi/2} (\sin\theta + t\cos\theta) f(t)dt$. Then the value of $\left|\int_0^{\pi/2} f(\theta)d\theta\right|$ is _____.

Open solution

Q41. Numerical

Let $\underset{0 \le x \le 2}{Max}\left\{\frac{9-x^2}{5-x}\right\} = \alpha$ and $\underset{0 \le x \le 2}{Min}\left\{\frac{9-x^2}{5-x}\right\} = \beta$.

If $\int\limits_{\beta-\frac{8}{3}}^{2\alpha-1} Max\left\{\frac{9-x^2}{5-x}, x\right\}dx = \alpha_1 + \alpha_2 \log_e\left(\frac{8}{15}\right)$ then $\alpha_1 + \alpha_2$ is equal to _____.

Open solution

Q42. Numerical

Let S be the region bounded by the curves $y = x^3$ and $y^2 = x$. The curve $y = 2|x|$ divides S into two regions of areas R_1, R_2. If max $\{R_1, R_2\} = R_2$, then $\frac{R_2}{R_1}$ is equal to _____.

Open solution

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Q43. Numerical

If the shortest distance between the lines

$\vec{r} = \left(-\hat{i} + 3\hat{k}\right) + \lambda\left(\hat{i} - a\hat{j}\right)$

and $\vec{r} = \left(-\hat{j} + 2\hat{k}\right) + \mu\left(\hat{i} - \hat{j} + \hat{k}\right)$ is $\sqrt{\frac{2}{3}}$, then the integral value of a is equal to _____.

Open solution

Physics 26 questions

Q44. MCQ (Single Correct Answer)

The bulk modulus of a liquid is 3×10^{10} Nm^{-2}. The pressure required to reduce the volume of liquid by 2% is :

(A) 3×10^{8} Nm^{-2} (B) 9×10^{8} Nm^{-2} (C) 6×10^{8} Nm^{-2} (D) 12×10^{8} Nm^{-2}

Open solution

Q45. MCQ (Single Correct Answer)

Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : In an uniform magnetic field, speed and energy remains the same for a moving charged particle.

Reason (R) : Moving charged particle experiences magnetic force perpendicular to its direction of motion.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

Open solution

Q46. MCQ (Single Correct Answer)

Two identical cells each of emf 1.5 V are connected in parallel across a parallel combination of two resistors each of resistance 20 Ω. A voltmeter connected in the circuit measures 1.2 V. The internal resistance of each cell is :

(A) 2.5 Ω (B) 4 Ω (C) 5 Ω (D) 10 Ω

Open solution

Q47. MCQ (Single Correct Answer)

Identify the pair of physical quantities which have different dimensions:

(A) Wave number and Rydberg's constant (B) Stress and Coefficient of elasticity

(C) Coercivity and Magnetisation (D) Specific heat capacity and Latent heat

Open solution

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Q48. MCQ (Single Correct Answer)

A projectile is projected with velocity of 25 m/s at an angle θ with the horizontal. After t seconds its inclination with horizontal becomes zero. If R represents horizontal range of the projectile, the value of θ will be :

[use g = 10 m/s^2]

(A) $\frac{1}{2}\sin^{-1}\left(\frac{5t^2}{4R}\right)$ (B) $\frac{1}{2}\sin^{-1}\left(\frac{4R}{5t^2}\right)$

(C) $\tan^{-1}\left(\frac{4t^2}{5R}\right)$ (D) $\cot^{-1}\left(\frac{R}{20t^2}\right)$

Open solution

Q49. MCQ (Single Correct Answer)

A block of mass 10 kg starts sliding on a surface with an initial velocity of 9.8 ms^{-1}. The coefficient of friction between the surface and block is 0.5. The distance covered by the block before coming to rest is :

[use g = 9.8 ms^{-2}]

(A) 4.9 m (B) 9.8 m (C) 12.5 m (D) 19.6 m

Open solution

Q50. MCQ (Single Correct Answer)

A boy ties a stone of mass 100 g to the end of a 2 m long string and whirls it around in a horizontal plane. The string can withstand the maximum tension of 80 N. If the maximum speed with which the stone can revolve is $\frac{K}{\pi}$ rev./min. The value of K is :

(Assume the string is massless and unstretchable)

(A) 400 (B) 300 (C) 600 (D) 800

Open solution

Q51. MCQ (Single Correct Answer)

A vertical electric field of magnitude 4.9×10^5 N/C just prevents a water droplet of a mass 0.1 g from falling. The value of charge on the droplet will be :

(Given : g = 9.8 m/s^2)

(A) 1.6×10^{-9} C (B) 2.0×10^{-9} C (C) 3.2×10^{-9} C (D) 0.5×10^{-9} C

Open solution

Q52. MCQ (Single Correct Answer)

A particle experiences a variable force $\vec{F} = \left(4x\hat{i} + 3y^2\hat{j}\right)$ in a horizontal x-y plane. Assume distance in meters and force is newton. If the particle moves from point (1, 2) to point (2, 3) in the x-y plane, then Kinetic Energy changes by :

(A) 50.0 J (B) 12.5 J (C) 25.0 J (D) 0 J

Open solution

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Q53. MCQ (Single Correct Answer)

The approximate height from the surface of earth at which the weight of the body becomes $\frac{1}{3}$ of its weight on the surface of earth is :

[Radius of earth R = 6400 km and $\sqrt{3}$ = 1.732]

Open solution

(A) 3840 km (B) 4685 km (C) 2133 km (D) 4267 km

Q54. MCQ (Single Correct Answer)

A resistance of 40 Ω is connected to a source of alternating current rated 220 V, 50 Hz. Find the time taken by the current to change from its maximum value to the rms value :

Open solution

(A) 2.5 ms (B) 1.25 ms (C) 2.5 s (D) 0.25 s

Q55. MCQ (Single Correct Answer)

The equations of two waves are given by :

$y_1 = 5 \sin 2\pi(x - vt)$ cm

$y_2 = 3 \sin 2\pi(x - vt + 1.5)$ cm

Open solution

These waves are simultaneously passing through a string. The amplitude of the resulting wave is :

(A) 2 cm (B) 4 cm (C) 5.8 cm (D) 8 cm

Q56. MCQ (Single Correct Answer)

A plane electromagnetic wave travels in a medium of relative permeability 1.61 and relative permittivity 6.44. If magnitude of magnetic intensity is 4.5×10^{-2} Am^{-1} at a point, what will be the approximate magnitude of electric field intensity at that point?

Open solution

(Given : Permeability of free space $\mu_0 = 4\pi \times 10^{-7}$ NA^{-2}, speed of light in vacuum $c = 3 \times 10^8$ ms^{-1})

(A) 16.96 Vm^{-1} (B) 2.25×10^{-2} Vm^{-1}

(C) 8.48 Vm^{-1} (D) 6.75×10^6 Vm^{-1}

Q57. MCQ (Single Correct Answer)

Choose the correct option from the following options given below :

Open solution

(A) In the ground state of Rutherford's model electrons are in stable equilibrium. While in Thomson's model electrons always experience a net-force.

(B) An atom has a nearly continuous mass distribution in a Rutherford's model but has a highly non-uniform mass distribution in Thomson's model.

(C) A classical atom based on Rutherford's model is doomed to collapse.

(D) The positively charged part of the atom possesses most of the mass in Rutherford's model but not in Thomson's model.

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Q58. MCQ (Single Correct Answer)

Nucleus A is having mass number 220 and its binding energy per nucleon is 5.6 MeV. It splits into two fragments 'B' and 'C' of mass numbers 105 and 115. The binding energy of nucleons in 'B' and 'C' is 6.4 MeV per nucleon. The energy Q released per fission will be :

Open solution

(A) 0.8 MeV (B) 275 MeV (C) 220 MeV (D) 176 MeV

Q59. MCQ (Single Correct Answer)

A parallel plate capacitor is formed by two plates each of area 30π cm^2 separated by 1 mm. A material of dielectric strength 3.6×10^7 Vm^{-1} is filled between the plates. If the maximum charge that can be stored on the capacitor without causing any dielectric breakdown is 7×10^{-6}C, the value of dielectric constant of the material is :

[Use $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9$ Nm^2 C^{-2}]

Open solution

(A) 1.66 (B) 1.75 (C) 2.25 (D) 2.33

Q60. MCQ (Single Correct Answer)

The magnetic field at the centre of a circular coil of radius r, due to current I flowing through it, is B. The magnetic field at a point along the axis at a distance $\frac{r}{2}$ from the centre is :

Open solution

(A) B/2 (B) 2B

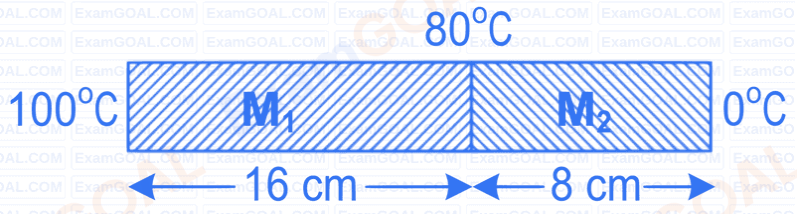
(C) $\left(\frac{2}{\sqrt{5}}\right)^3 B$ (D) $\left(\frac{2}{\sqrt{3}}\right)^3 B$

Q61. MCQ (Single Correct Answer)

Two metallic blocks M_1 and M_2 of same area of cross-section are connected to each other (as shown in figure). If the thermal conductivity of M_2 is K then the thermal conductivity of M_1 will be :

[Assume steady state heat conduction]

Open solution



(A) 10 K (B) 8 K (C) 12.5 K (D) 2 K

Q62. Numerical

0.056 kg of Nitrogen is enclosed in a vessel at a temperature of 127°C. The amount of heat required to double the speed of its molecules is _____ k cal.

(Take R = 2 cal mole^{-1} K^{-1})

Open solution

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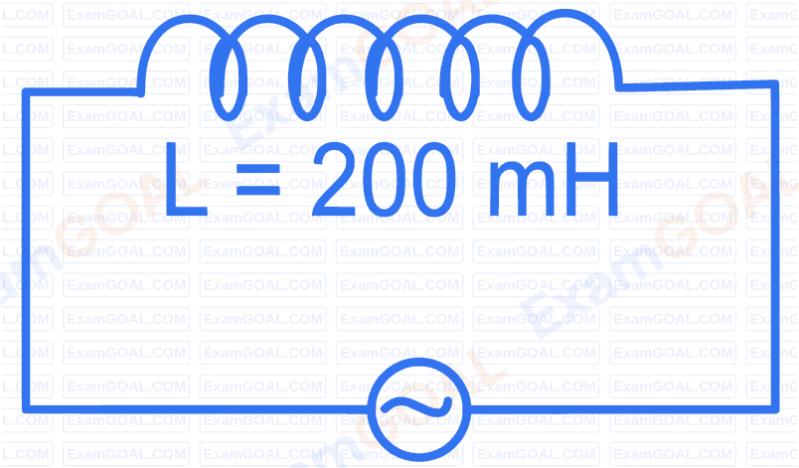
Q63. Numerical

Two identical thin biconvex lens of focal length 15 cm and refractive index 1.5 are in contact with each other. The space between the lenses is filled with a liquid of refractive index 1.25. The focal length of the combination is _____ cm.

Open solution

Q64. Numerical

As shown in the figure an inductor of inductance 200 mH is connected to an AC source of emf 220 V and frequency 50 Hz. The instantaneous voltage of the source is 0 V when the peak value of current is $\frac{\sqrt{a}}{\pi}$ A. The value of a is _____.



Open solution

Q65. Numerical

Sodium light of wavelengths 650 nm and 655 nm is used to study diffraction at a single slit of aperture 0.5 mm. The distance between the slit and the screen is 2.0 m. The separation between the positions of the first maxima of diffraction pattern obtained in the two cases is _____ $\times 10^{-5}$ m.

Open solution

Q66. Numerical

When light of frequency twice the threshold frequency is incident on the metal plate, the maximum velocity of emitted electron is v_1. When the frequency of incident radiation is increased to five times the threshold value, the maximum velocity of emitted electron becomes v_2. If $v_2 = x\ v_1$, the value of x will be _____.

Open solution

Q67. Numerical

From the top of a tower, a ball is thrown vertically upward which reaches the ground in 6 s. A second ball thrown vertically downward from the same position with the same speed reaches the ground in 1.5 s. A third ball released, from the rest from the same location, will reach the ground in _____ s.

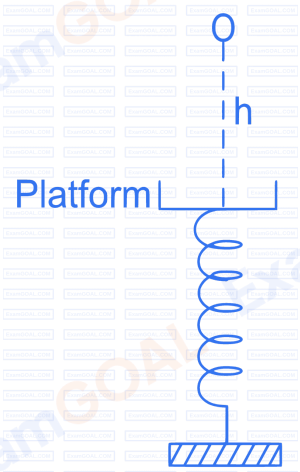
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Q68. Numerical

A ball of mass 100 g is dropped from a height h = 10 cm on a platform fixed at the top of a vertical spring (as shown in figure). The ball stays on the platform and the platform is depressed by a distance $\frac{h}{2}$. The spring constant is _____ Nm^{-1}.

(Use g = 10 ms^{-2})



Open solution

Q69. Numerical

A metre scale is balanced on a knife edge at its centre. When two coins, each of mass 10 g are put one on the top of the other at the 10.0 cm mark the scale is found to be balanced at 40.0 cm mark. The mass of the metre scale is found to be x $\times$ 10^{-2} kg. The value of x is _____.

Open solution

Out of syllabus 21 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q70. MCQ (Single Correct Answer) · **Chemistry**

Given below are two statements :

Statement I : Emulsions of oil in water are unstable and sometimes they separate into two layers on standing.

Statement II : For stabilisation of an emulsion, excess of electrolyte is added.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

Open solution

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Q71. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Match List - I with List - II :

	List - I		List -II
(A)	Sphalerite	(I)	$FeCO_3$
(B)	Calamine	(II)	PbS
(C)	Galena	(III)	$ZnCO_3$
(D)	Siderite	(IV)	ZnS

Choose the most appropriate answer from the options given below :

(A) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(B) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

(C) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

(D) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

Q72. MCQ (Single Correct Answer) · **Chemistry**

Open solution

The highest industrial consumption of molecular hydrogen is to produce compounds of element :

(A) Carbon **(B)** Nitrogen **(C)** Oxygen **(D)** Chlorine

Q73. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Which of the following statements are correct?

(A) Both LiCl and $MgCl_2$ are soluble in ethanol.

(B) The oxides Li_2O and MgO combine with excess of oxygen to give superoxide.

(C) LiF is less soluble in water than other alkali metal fluorides.

(D) Li_2O is more soluble in water than other alkali metal oxides.

Choose the most appropriate answer from the options given below :

(A) (A) and (C) only

(B) (A), (C) and (D) only

(C) (B) and (C) only

(D) (A) and (D) only

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Q74. MCQ (Single Correct Answer) · **Chemistry**

Identify the correct statement for B_2H_6 from those given below :

(A) In B_2H_6, all B-H bonds are equivalent.

(B) In B_2H_6, there are four 3-centre-2-electron bonds.

(C) B_2H_6 is a Lewis acid.

(D) B_2H_6 can be synthesized from both BF_3 and $NaBH_4$.

(E) B_2H_6 is a planar molecule.

Choose the most appropriate answer from the options given below :

(A) (A) and (E) only

(B) (B), (C) and (E) only

(C) (C) and (D) only

(D) (C) and (E) only

Open solution

Q75. MCQ (Single Correct Answer) · **Chemistry**

Which one of the following elemental forms is not present in the enamel of the teeth?

(A) Ca^{2+} **(B)** P^{3+} **(C)** F^- **(D)** P^{5+}

Open solution

Q76. MCQ (Single Correct Answer) · **Chemistry**

Which of the following is an example of polyester?

(A) Butadiene-styrene copolymer

(B) Melamine polymer

(C) Neoprene

(D) Poly-β-hydroxybutyrate-co-β-hydroxyvalerate

Open solution

Q77. MCQ (Single Correct Answer) · **Chemistry**

Which of the following is not a broad spectrum antibiotic?

(A) Vancomycin **(B)** Ampicillin **(C)** Ofloxacin **(D)** Penicillin G

Open solution

Q78. Numerical · **Chemistry**

Atoms of element X form hcp lattice and those of element Y occupy $\frac{2}{3}$ of its tetrahedral voids. The percentage of element X in the lattice is _____. (Nearest integer)

Open solution

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Q79. MCQ (Single Correct Answer) · **Mathematics**

Let $x^2 + y^2 + Ax + By + C = 0$ be a circle passing through (0, 6) and touching the parabola $y = x^2$ at (2, 4). Then A + C is equal to _____.

(A) 16 (B) 88/5 (C) 72 (D) -8

Open solution

Q80. MCQ (Single Correct Answer) · **Mathematics**

Let S = $\{\sqrt{n} : 1 \le n \le 50$ and n is odd$\}$.

Let a $\in$ S and $A = \begin{bmatrix} 1 & 0 & a \\ -1 & 1 & 0 \\ -a & 0 & 1 \end{bmatrix}$.

If $\sum_{a \in S} \det(adj\, A) = 100\lambda$, then λ is equal to :

(A) 218 (B) 221 (C) 663 (D) 1717

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

If the tangent at the point (x_1, y_1) on the curve $y = x^3 + 3x^2 + 5$ passes through the origin, then (x_1, y_1) does NOT lie on the curve :

(A) $x^2 + \frac{y^2}{81} = 2$ (B) $\frac{y^2}{9} - x^2 = 8$ (C) $y = 4x^2 + 5$ (D) $\frac{x}{3} - y^2 = 2$

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

Let $\lambda x - 2y = \mu$ be a tangent to the hyperbola $a^2x^2 - y^2 = b^2$. Then $\left(\frac{\lambda}{a}\right)^2 - \left(\frac{\mu}{b}\right)^2$ is equal to :

(A) -2 (B) -4 (C) 2 (D) 4

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

If a random variable X follows the Binomial distribution B(33, p) such that

$3P(X = 0) = P(X = 1)$, then the value of $\frac{P(X=15)}{P(X=18)} - \frac{P(X=16)}{P(X=17)}$ is equal to :

(A) 1320 (B) 1088 (C) $\frac{120}{1331}$ (D) $\frac{1088}{1089}$

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

Let $S = \left\{\theta \in [-\pi, \pi] - \left\{\pm \frac{\pi}{2}\right\} : \sin\theta\tan\theta + \tan\theta = \sin 2\theta\right\}$.

If $T = \sum_{\theta \in S} \cos 2\theta$, then T + n(S) is equal to :

(A) $7 + \sqrt{3}$ (B) 9 (C) $8 + \sqrt{3}$ (D) 10

Open solution

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Q85. MCQ (Single Correct Answer) · **Mathematics**

The number of choices for $\Delta \in \{\wedge, \vee, \Rightarrow, \Leftrightarrow\}$, such that

$(p\Delta q) \Rightarrow ((p\Delta \sim q) \vee ((\sim p)\Delta q))$ is a tautology, is :

(A) 1 (B) 2 (C) 3 (D) 4

Open solution

Q86. Numerical · **Mathematics**

If two tangents drawn from a point (α, β) lying on the ellipse $25x^2 + 4y^2 = 1$ to the parabola $y^2 = 4x$ are such that the slope of one tangent is four times the other, then the value of $(10\alpha + 5)^2 + (16\beta^2 + 50)^2$ equals _____.

Open solution

Q87. MCQ (Single Correct Answer) · **Physics**

A baseband signal of 3.5 MHz frequency is modulated with a carrier signal of 3.5 GHz frequency using amplitude modulation method. What should be the minimum size of antenna required to transmit the modulated signal?

(A) 42.8 m (B) 42.8 mm (C) 21.4 mm (D) 21.4 m

Open solution

Q88. MCQ (Single Correct Answer) · **Physics**

A Carnot engine whose heat sink is at 27°C, has an efficiency of 25%. By how many degrees should the temperature of the source be changed to increase the efficiency by 100% of the original efficiency?

(A) Increases by 18°C (B) Increases by 200°C

(C) Increases by 120°C (D) Increases by 73°C

Open solution

Q89. Numerical · **Physics**

A transistor is used in common-emitter mode in an amplifier circuit. When a signal of 10 mV is added to the base-emitter voltage, the base current changes by 10 μA and the collector current changes by 1.5 mA. The load resistance is 5 kΩ. The voltage gain of the transistor will be _____.

Open solution

Q90. Numerical · **Physics**

In a potentiometer arrangement, a cell gives a balancing point at 75 cm length of wire. This cell is now replaced by another cell of unknown emf. If the ratio of the emf's of two cells respectively is 3 : 2, the difference in the balancing length of the potentiometer wire in above two cases will be _____ cm.

Open solution

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Answer Key

Chemistry

1. **C**	2. **B**	3. **B**	4. **B**	5. **B**
6. **A**	7. **B**	8. **A**	9. **C**	10. **D**
11. **B**	12. **A**	13. **747**	14. **54**	15. **5**
16. **154**	17. **0**	18. **7**	19. **63**	20. **3**
21. **2**				

Mathematics

22. **D**	23. **D**	24. **A**	25. **C**	26. **B**
27. **B**	28. **A**	29. **C**	30. **B**	31. **B**
32. **A**	33. **C**	34. **D**	35. **31**	36. **40**
37. **8**	38. **84**	39. **7**	40. **1**	41. **34**
42. **19**	43. **2**			

Physics

44. **C**	45. **A**	46. **C**	47. **D**	48. **D**
49. **B**	50. **C**	51. **B**	52. **C**	53. **B**
54. **A**	55. **A**	56. **C**	57. **C**	58. **D**
59. **D**	60. **C**	61. **B**	62. **12**	63. **10**
64. **242**	65. **3**	66. **2**	67. **3**	68. **120**
69. **6**				

Out of syllabus

70. **C**	71. **A**	72. **B**	73. **A**	74. **C**
75. **B**	76. **D**	77. **D**	78. **43**	79. **A**
80. **B**	81. **D**	82. **D**	83. **A**	84. **B**
85. **B**	86. **2929**	87. **C**	88. **B**	89. **750**
90. **25**				

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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